Lingnan University Intelligent Consultation System Development Plan

1. Executive Summary

This plan outlines the full lifecycle development of an Intelligent Consultation System for Lingnan University, aiming to solve the inefficiency of information retrieval for students and faculty regarding university policy documents, course guidelines, and administrative matters. The system is built on the AgenticRAG (Retrieval-Augmented Generation with Agent Orchestration) framework, integrating semantic retrieval, generative AI, and role-based access control to provide accurate, efficient, and user-friendly information consultation services. This document covers project background, pain points, objectives, feasibility analysis, system design, implementation plan, resource allocation, risk management, and expected benefits, serving as a comprehensive guide for the project's smooth promotion and implementation.

2. Project Background

2.1 Institutional Context

Lingnan University, as a prestigious institution of higher education, adheres to the concept of "student-oriented, faculty-driven" and attaches great importance to improving the service quality of teaching and administration. With the continuous expansion of the university's scale, the enrichment of academic programs, and the refinement of administrative management, many policy documents, course materials, and administrative guidelines have been formed, involving academic management, student services, faculty support, financial management, and other fields.

These documents are the basis for standardizing the university's operation and safeguarding the legitimate rights and interests of students and faculty. However, with the increase in document quantity and the frequent update of content, the traditional information retrieval method has gradually failed to meet the actual needs of users, becoming a bottleneck restricting the improvement of the university's service efficiency and user experience.

2.2 Industry Trends

In recent years, with the rapid development of artificial intelligence, especially natural language processing (NLP) and large language models (LLMs), intelligent consultation systems have been widely applied in education, government, enterprises and other fields. Universities at home and abroad have begun to explore the use of AI technology to optimize information services, reduce administrative costs, and improve user satisfaction. For example, some well-known universities have launched intelligent customer service systems to answer students' questions about enrollment, course selection, and graduation, achieving good results in improving service efficiency and reducing manual workload.

Against this background, Lingnan University's development of an intelligent

consultation system is not only in line with the trend of educational digital transformation but also an important measure to promote the university's high-quality development and enhance its core competitiveness.

3. Pain Points Analysis

Through investigation and research on students, faculty, and administrative staff of Lingnan University, the main pain points in the current information retrieval process are summarized as follows:

3.1 For Students

- ✧ **Dispersed Document Sources**: University policy documents, course syllabuses, and administrative guidelines are scattered across multiple platforms (such as the university official website, departmental homepages, and learning management systems), making it difficult for students to quickly locate the required information.
- ✧ **Inefficient Retrieval Process**: Traditional keyword search can only match literal content, and it is impossible to understand the user's semantic intent. Students often need to browse many irrelevant documents to find accurate answers, which is time-consuming and labor-intensive.
- ✧ **Unclear Information Understanding**: Some policy documents are professional and lengthy, and students may have difficulty accurately understanding the core content, leading to confusion in handling related matters (such as course registration, academic integrity, and financial aid).
- ✧ **Limited Consultation Channels**: The traditional consultation method relies on administrative office hours, and students cannot get timely answers to urgent questions outside of working hours.

3.2 For Faculty

- ✧ **Repetitive Workload**: Faculty often receive many repetitive questions from students (such as course requirements, assignment submission deadlines, and academic evaluation standards), which takes up a lot of time and energy that could be used for teaching and scientific research.
- ✧ **Difficulty in Information Synchronization**: When university policies or course guidelines are updated, faculty need to manually inform students one by one, which is inefficient and prone to information asymmetry.
- ✧ **Lack of Centralized Management**: Course materials and assignment guidelines are stored in different forms (such as local files, email attachments), making it difficult for faculty to manage and update them uniformly.

3.3 For Administrative Staff

- ✧ **Heavy Consulting Workload**: Administrative staff need to answer many user inquiries every day, resulting in high work pressure and difficulty in ensuring the consistency and accuracy of answers.
- ✧ **Difficulty in Data Statistics**: It is difficult to track and count the types and frequencies of user inquiries, making it impossible to accurately grasp the needs

of students and faculty and optimize policies and services in a targeted manner.

- ✧ **High Cost of Document Maintenance**: With the frequent update of policy documents, administrative staff need to manually update the content on multiple platforms, which is inefficient and prone to errors.

4. Project Objectives

4.1 Overall Objective

Develop a professional, efficient, and intelligent consultation system tailored to Lingnan University, which can accurately answer users' questions about university policies, courses, and administrative matters, simplify the information retrieval process, reduce the workload of faculty and administrative staff, and improve the satisfaction of students and faculty with university services.

4.2 Specific Objectives

- ✧ **Accuracy**: The system's answer accuracy rate for standard questions (such as policy inquiries) reaches more than 95%, ensuring that the answers are based on official university documents and avoiding misinformation.
- ✧ **Efficiency**: Reduce the average information retrieval time for users by more than 70%, and the system response time is within 10 seconds, realizing real-time consultation.
- ✧ **User Experience**: Provide an intuitive and user-friendly operation interface, support natural language dialogue, maintain conversation context, and meet the personalized needs of different users (students, faculty, administrative staff).
- ✧ **Scalability**: The system architecture is designed to be scalable, supporting the subsequent addition of new functions (such as multilingual support, voice consultation) and the integration with other university information systems (such as student information system, faculty management system).
- ✧ **Compliance**: Comply with the university's data security and privacy protection policies, implement role-based access control, and ensure the security and confidentiality of user information and official documents.

5. Feasibility Analysis

5.1 Technical Feasibility

The project adopts mature and reliable technical frameworks and tools, which have been verified in practical applications and have high technical feasibility:

- ✧ **AgenticRAG Framework**: As an advanced RAG solution, it combines agent orchestration, semantic retrieval, and generative AI, which can effectively solve the problem of inaccurate answers and improve the intelligence of the system. The framework has open-source code and mature technical documentation, which is convenient for secondary development and customization.
- ✧ **Technology Stack**: The front-end adopts React + TypeScript + Vite, which has the advantages of high efficiency, good compatibility, and easy maintenance; the

back-end adopts FastAPI, which is fast and easy to integrate with AI models; the database adopts PostgreSQL + pgvector, which supports vector storage and semantic retrieval, ensuring the efficiency and accuracy of data retrieval; the LLM adopts Google Vertex AI (Gemini) and local Ollama as fallback, ensuring the stability and availability of the system.

- ✧ **Technical Team**: The project team has rich experience in AI development, software engineering, and educational informationization, and can provide technical support for the project's development, deployment, and maintenance.

5.2 Economic Feasibility

The project's investment is mainly concentrated in the development, deployment, and maintenance stages, and the long-term economic benefits are significant:

- ✧ **Reduced Labor Costs**: The system can replace most of the manual consultation work, reducing the workload of administrative staff and faculty, and saving labor costs for the university.
- ✧ **Improved Work Efficiency**: By simplifying the information retrieval process, the work efficiency of students, faculty, and administrative staff is improved, which helps to promote the smooth development of the university's teaching and administrative work.
- ✧ **Low Maintenance Costs**: The system adopts a modular and decoupled architecture, which is easy to maintain and update. The maintenance cost after deployment is relatively low, and the investment return cycle is short.

5.3 Operational Feasibility

The system is designed to be user-friendly and in line with the usage habits of students and faculty, with high operational feasibility:

- ✧ **User-Friendly Interface**: The front-end interface is simple and intuitive, supporting natural language dialogue, and users can complete consultation without professional technical knowledge.
- ✧ **Integration with Existing Systems**: The system can be integrated with the university's existing authentication system (such as LDAP/SSO), avoiding duplicate account creation and facilitating user login and use.
- ✧ **Training Support**: The project team will provide training sessions for users and administrative staff, helping them quickly master the use and maintenance methods of the system.

5.4 Policy Feasibility

The project is in line with the national policy of promoting educational digital transformation and the university's strategy of improving service quality. The university attaches great importance to the application of information technology in teaching and administration and will provide policy support and resource guarantee for the project's implementation.

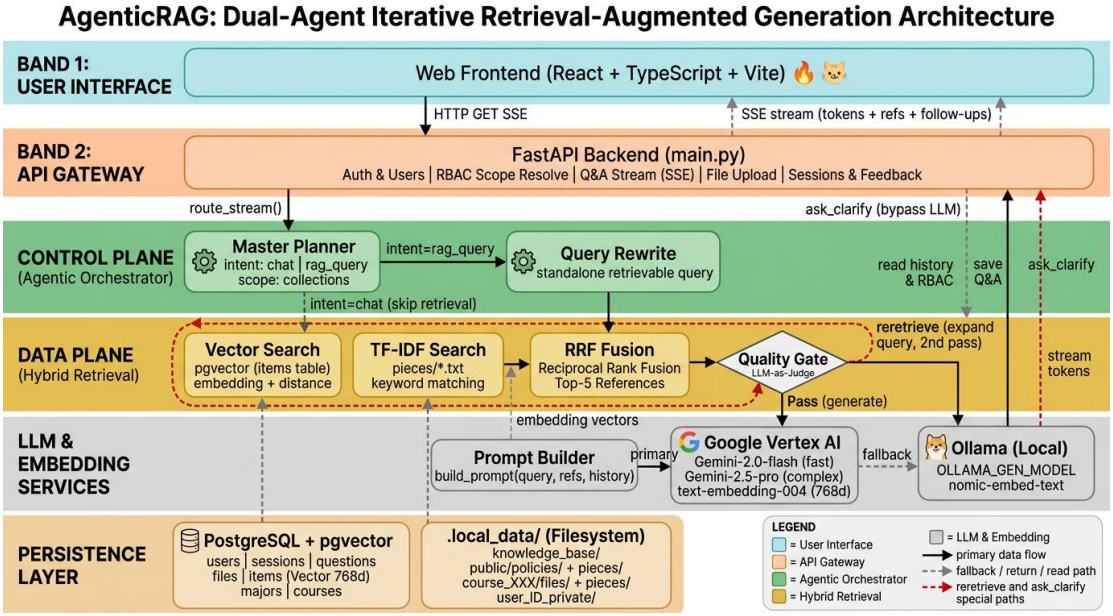
6. System Design

6.1 Design Principles

- ✧ **User-Centered**: Taking the needs of students, faculty, and administrative staff as the core, optimizing the user interface and interaction process to improve user experience.
- ✧ **Accuracy and Reliability**: Grounding the system's answers in official university documents to ensure the accuracy and authority of the information.
- ✧ **Scalability and Maintainability**: Adopting a modular and decoupled architecture, facilitating the addition of new functions and the maintenance of the system.
- ✧ **Security and Compliance**: Implementing strict data security and privacy protection measures to comply with the university's relevant policies and regulations.

6.2 System Architecture

The system adopts a decoupled front-end/back-end architecture, which is divided into six core layers to ensure the scalability, maintainability, and security of the system. The specific architecture is as follows:



6.2.1 Front-End Layer (User Interface)

Technology Stack: React + TypeScript + Vite

Key Functions:

- ✧ Intuitive chat interface for natural language Q&A, supporting streaming responses to reduce user wait time.
- ✧ File management module, supporting users to upload, browse, and delete policy documents, course materials, etc.
- ✧ User authentication portal, supporting login through the university's existing SSO system.
- ✧ Session history view, allowing users to revisit past queries and answers.

- ✧ Responsive design, adapting to desktop, mobile phones, and other devices to facilitate users to consult anytime and anywhere.

6.2.2 Back-End API Layer (Service Orchestration)

Technology Stack: FastAPI (Python)

Key Functions:

- ✧ Providing RESTful API endpoints for user authentication, file management, Q&A interaction, and other functions.
- ✧ Agent orchestration, including query intent recognition, query rewriting, and result integration, to improve the accuracy of retrieval and generation.
- ✧ Integrating with the retrieval layer, LLM layer, and database layer to realize the coordination and interaction of various modules.
- ✧ Providing API documentation (Swagger UI) to facilitate the maintenance and debugging of the internal IT team.

6.2.3 Retrieval Layer (Hybrid Data Retrieval)

To balance the precision and recall of retrieval, the system adopts a hybrid retrieval method with RRF (Reciprocal Rank Fusion) fusion:

- ✧ **Semantic Retrieval**: Using pgvector (PostgreSQL extension) to store and search vector embeddings of document content, realizing semantic understanding and matching of queries.
- ✧ **Keyword Retrieval**: Using TF-IDF algorithm to search text fragments, ensuring that literal matching queries can also get accurate results.
- ✧ **Fusion Logic**: Combining the top 5 results of semantic retrieval and keyword retrieval to ensure comprehensive coverage of relevant documents and improve retrieval accuracy.
- ✧ **Quality Gate**: Following RRF fusion, the system applies an LLM-as-Judge checkpoint that evaluates whether the top-5 retrieved references are sufficient to answer the user's question rigorously. If the references are adequate, the system proceeds to generation; if insufficient, it automatically expands the query with domain-specific keywords and performs a second retrieval pass, comparing both result sets and selecting the higher-quality references; if retrieval remains insufficient after retry, the system returns a targeted clarifying question to the user along with the matched source files, preserving transparency and guiding them toward a more precise query.

6.2.4 LLM/Embedding Layer (Generation & Semantic Encoding)

- ✧ **Primary LLM**: Google Vertex AI (Gemini), which has the advantages of enterprise-grade reliability, multilingual support (English/Cantonese), and strong generation ability, suitable for the university's bilingual environment.
- ✧ **Fallback LLM**: Local Ollama (such as Llama 3.2), which is deployed locally to ensure that the system can still work normally when the cloud-based Vertex AI is unavailable.
- ✧ **Embedding Model**: Vertex AI text-embedding-004, which converts documents

and queries into semantic vectors, providing support for semantic retrieval.

6.2.5 Persistence Layer (Data Storage)

- ✧ **Database**: PostgreSQL + pgvector, which stores user data (accounts, roles), file metadata, session history, Q&A records, and vector embeddings. It supports ACID compliance to ensure data integrity and security.
- ✧ **File Storage**: Local directory (`local_data/`) with subdirectories for knowledge base, user uploads, and avatars. It can be configured to cloud storage (such as Google Cloud Storage) according to the university's data governance requirements.

6.2.6 Security & Access Control Layer

Implementing role-based access control (RBAC) to ensure that different users have different operation permissions:

- ✧ **Students**: Can access public policy documents, course materials, and their own Q&A history, and ask questions about relevant content.
- ✧ **Faculty**: In addition to the permissions of students, they can also access course assignment guidelines and department-specific policies and manage their own course materials.
- ✧ **Admins**: Have the highest permissions, including file upload/management, system configuration, user role management, and audit logs.

6.3 Core Functions

6.3.1 User Management

- ✧ Seamless integration with the university's existing authentication system (LDAP/SSO) to avoid duplicate account creation.
- ✧ User role assignment, including students, faculty, and admins, aligned with the university's organizational structure.
- ✧ User information management, supporting the modification of personal information and password.
- ✧ Audit logs, recording the operation behavior of admins (such as file uploads, user role changes) to ensure compliance.

6.3.2 File Management

- ✧ Supporting the upload of common document formats (PDF, DOCX, TXT) for policy documents, course syllabuses, and assignment guidelines.
- ✧ Automatic indexing of documents: After uploading, documents are split into text fragments, converted into vector embeddings, and stored in the database for retrieval.
- ✧ Document management: Supporting the browsing, editing, and deletion of documents, and ensuring that deleting a document will also delete its

corresponding text fragments and vector embeddings to avoid stale data.

- ✧ Document classification: Supporting the classification of documents according to categories (such as academic management, student services) to facilitate retrieval.

6.3.3 Intelligent Q&A

- ✧ Natural language dialogue: Supporting users to ask questions in natural language, and the system can understand the semantic intent and give accurate answers.
- ✧ Context awareness: Maintaining the conversation history, supporting follow-up questions, and providing continuous consultation services.
- ✧ Streaming responses: Answers are generated in real time (token-by-token) to reduce user wait time.
- ✧ Temporary file uploads: Users can upload specific documents (such as course syllabuses) to ask targeted questions about the content of the documents.
- ✧ Answer feedback: Allowing users to rate the accuracy of answers (thumbs up/down) to provide a basis for system optimization.

6.3.4 Data Statistics & Analysis

- ✧ Query statistics: Counting the types, frequencies, and satisfaction of user queries to grasp user needs.
- ✧ System performance statistics: Monitoring the system's response time, retrieval accuracy, and LLM fallback rate to identify bottlenecks.
- ✧ Report generation: Generating regular statistics reports to provide decision support for university administrators.

7. Implementation Plan

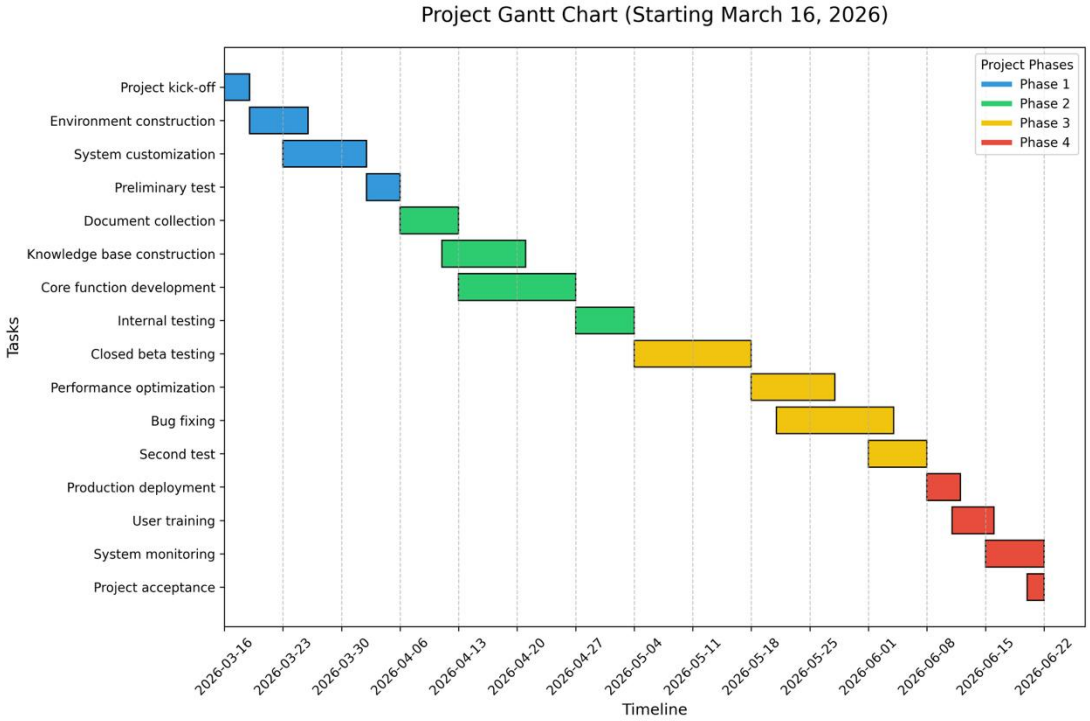
7.1 Project Team Composition

Role	Number of People	Responsibilities
Project Manager	1	Overall responsible for the project, coordinating resources, controlling progress, and handling project risks.
Front-End Developer	2	Responsible for the development and optimization of the front-end interface and user interaction.
Back-End Developer	2	Responsible for the development of back-end APIs, agent orchestration, and integration with other modules.
AI Engineer	2	Responsible for the integration of LLM

		and embedding models, and the optimization of retrieval and generation effects.
Database Engineer	1	Responsible for the design, deployment, and maintenance of the database.
Test Engineer	1	Responsible for the testing of the system, including functional testing, performance testing, and security testing.
User Trainer	1	Responsible for preparing training materials and providing training sessions for users and admins.

7.2 Phased Implementation Plan

The project is divided into four phases, with a total cycle of 11–14 weeks, ensuring the smooth implementation and launch of the system.



7.3 Key Milestones

Milestone	Time Node	Deliverables
Project kick-off and environment construction completed	Week 3	Development environment, customized front-end UI, integrated authentication system

Core function development and knowledge base construction completed	Week 7	Functional system, initial knowledge base, internal test report
User testing and system optimization completed	Week 12	Optimized system, beta test report, feedback summary
Full deployment and training completed	Week 14	Production system, training materials, user manual, project acceptance report

8. Resource Allocation

8.1 Human Resources

As shown in Section 7.1, the project team consists of 10 people, including project manager, front-end developers, back-end developers, AI engineers, database engineer, test engineer, and user trainer. The university needs to assign special personnel to cooperate with the project team in document collection, user organization, and project acceptance.

8.2 Hardware Resources

- **Development Server**: 2 servers (CPU: Intel Xeon E5-2690, RAM: 32GB, HDD: 1TB) for deploying the development environment and testing environment.
- **Production Server**: 2 servers (CPU: Intel Xeon Gold 6248, RAM: 64GB, HDD: 2TB) for deploying the production system, ensuring high availability.
- **Database Server**: 1 server (CPU: Intel Xeon Gold 6248, RAM: 64GB, HDD: 4TB) for deploying PostgreSQL database, ensuring data storage and retrieval efficiency.

8.3 Software Resources

- **Development Software**: VS Code, PyCharm, Figma (for UI design), Docker, Docker Compose.
- **Operating System**: Ubuntu Server 20.04 LTS (for servers), Windows 11/macOS (for development computers).
- **AI Services**: Google Vertex AI (Gemini), Ollama (local LLM).
- **Database Software**: PostgreSQL 15, pgvector extension.

8.4 Budget Estimation

Budget Category	Estimated Cost (HKD)	Description
Human Resource Cost	350,000–400,000	Salaries of project team members during the project period

Hardware Cost	200,000–250,000	Purchase of development and production servers, database servers
Software and Service Cost	50,000–80,000	AI service fees, software licenses, etc.
Training Cost	20,000–30,000	Preparation of training materials, training sessions
Contingency Cost	30,000–40,000	Used to deal with unexpected situations during the project
Total	650,000–800,000	—

9. Risk Management

9.1 Risk Identification & Mitigation Strategies

Risk Category	Specific Risk	Impact Level	Mitigation Strategy
Technical Risk	LLM hallucinations lead to inaccurate answers	High	Strict prompt engineering to enforce grounding in retrieved documents; set up a human review mechanism for high-stakes queries (such as financial aid, academic punishment); regularly optimize LLM prompts and retrieval parameters based on user feedback.
Technical Risk	System downtime or poor performance	Medium	Deploy redundant back-end and LLM instances; use Ollama as a local fallback for Vertex AI; regularly back up the database; set up real-time monitoring to find and solve problems in time.
Resource Risk	Insufficient hardware/software resources affect project progress	Medium	Formulate a detailed resource allocation plan in advance; communicate with the university to ensure the timely supply of resources; prepare alternative resources (such as

			cloud servers) in case of emergency.
Management Risk	Project progress delays due to unclear division of responsibilities	Medium	Clarify the division of responsibilities of each team member at the beginning of the project; hold weekly project progress meetings to track progress and solve problems in time; set up milestone checks to ensure the project is completed on schedule.
User Risk	Low user adoption rate	High	Optimize the UI/UX to make the system easy to use; conduct extensive training and promotion through university channels (email, official website, departmental notices); collect user feedback and continuously improve the system.
Security Risk	Data security and privacy leaks	High	Encrypt sensitive data (user accounts, documents); implement role-based access control to restrict user permissions; regularly update system dependencies to patch vulnerabilities; conduct security testing to find and fix security risks.

9.2 Risk Monitoring & Response

The project manager is responsible for risk monitoring, and the team holds weekly risk assessment meetings to track the status of identified risks and identify new risks. For each risk, a special person is assigned to be responsible for implementing the mitigation strategy. If a risk occurs, the responsible person will immediately take response measures to minimize the impact on the project.

10. Maintenance & Operation Plan

10.1 Daily Maintenance

- **System Monitoring**: Monitor the system's running status, including API response time, database performance, and LLM availability, and deal with abnormal situations in time.

- **Data Backup**: Back up the database and file storage daily to ensure data security. The backup data is stored in multiple locations to prevent data loss.
- **Bug Fixing**: Collect user feedback and system logs, find and fix bugs in a timely manner, and release patch updates.

10.2 Regular Updates

- **Knowledge Base Update**: Cooperate with the university's administrative departments to regularly update the policy documents and course materials in the knowledge base to ensure that the information is up to date.
- **System Function Update**: According to user needs and industry trends, regularly add new functions (such as multilingual support, voice consultation) and optimize existing functions.
- **Model Optimization**: Retrain the embedding model and optimize LLM prompts regularly to improve the accuracy and quality of answers.

10.3 Operation Support

Set up a dedicated operation support team to answer user questions and solve user problems in a timely manner. Regularly collect user feedback and conduct user satisfaction surveys to continuously improve the system. Provide technical support for the university's IT team to help them master the maintenance methods of the system.

11. Expected Outcomes & Benefits

11.1 For Students

Efficient Information Retrieval: Reduce the time spent on information retrieval by more than 70%, and quickly obtain accurate answers to questions about policies, courses, and administrative matters. Improved Learning Experience: Clear and concise answers help students better understand policy requirements and course guidelines, reducing confusion and improving learning efficiency. 24/7 Consultation Service: The system is available 24 hours a day, 7 days a week, allowing students to get timely answers to urgent questions outside of working hours.

11.2 For Faculty

Reduced Workload: The system answers most of the repetitive questions from students, freeing up faculty's time and energy for teaching and scientific research. Efficient Information Synchronization: When policies or course guidelines are updated, faculty can quickly update the content in the system, ensuring that students receive the latest information in a timely manner. Centralized Document Management: Course materials and assignment guidelines can be managed uniformly in the system, improving management efficiency.

11.3 For University Administrators

Improved Administrative Efficiency: Reduce the workload of administrative staff, simplify the consultation process, and improve the efficiency of administrative work. Data-Driven Decision Making: Through the statistics and analysis of user queries,

grasp the needs of students and faculty, and provide decision support for policy optimization and service improvement. Enhanced Service Quality: The system provides accurate, efficient, and user-friendly consultation services, improving the satisfaction of students and faculty with university services, and enhancing the university's image.

11.4 For the University

Promote Digital Transformation: The system is an important practice of the university's educational digital transformation, helping the university build a smart campus. Enhance Core Competitiveness: By improving service quality and work efficiency, the university's core competitiveness is enhanced, which is conducive to attracting students and faculty. Standardize Information Management: The system centralizes the management of policy documents and course materials, ensuring the standardization and authority of information.

12.Future

13. Conclusion

The development of the Intelligent Consultation System for Lingnan University is a necessary measure to solve the current pain points of information retrieval, improve service quality, and promote digital transformation. The system, based on the AgenticRAG framework, integrates advanced AI technologies and is tailored to the actual needs of the university, with high technical feasibility, economic feasibility, and operational feasibility.

Through the phased implementation plan, the project will be completed smoothly, and the system will be put into use to provide accurate, efficient, and user-friendly consultation services for students, faculty, and administrative staff. The system will not only reduce the workload of relevant personnel but also improve the satisfaction of users with university services, providing strong support for the university's high-quality development.

The project team will strictly follow the plan to promote the project, strengthen risk management, ensure the quality and progress of the project, and deliver a high-quality intelligent consultation system to Lingnan University.